Chapter 5. Approximating Sampling Distributions

Central limit theorem, convergence in distribution and probability, limit computations, simulation

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Aim

Beyond normal samples (Chapter 4), sampling distributions are approximated by limits (CLT, LLN, Slutsky, delta method) or by simulation.

5.1 The central limit theorem

Already in Chapter 3 we saw the normal approximation to the binomial. If $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$ and $\hat{p} = \bar{X}_n$, then

$$\lim_{n\to\infty} P\Big(a \le \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \le b\Big) = P(a \le Z \le b), \qquad Z \sim N(0,1).$$

The central limit theorem says that the same holds for the sample mean of any population with finite variance.

Theorem 5.1 (Central limit theorem)

Let $X_1, \dots, X_n$ be i.i.d. with $EX_1 = \mu$ and $\operatorname{Var} X_1 = \sigma^2 \in (0, \infty)$. Then for every x,

$$\lim_{n\to\infty} P\Big(\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \le x\Big) = \Phi(x).$$

Proof. We assume additionally an mgf near 0 and use the *continuity theorem*: if $M_{Z_n}(t) \to M_Z(t)$ for all $|t| < h$ and Z has a continuous cdf, then $P(Z_n \le x) \to P(Z \le x)$ for all x.

Let $m(s)$ be the mgf of $(X_1 - \mu)/\sigma$, so $m(0) = 1$, $m'(0) = 0$, $m''(0) = 1$. Since the summands are i.i.d.,

$$M_{\sqrt{n}(\bar{X}_n - \mu)/\sigma}(t) = E \exp\Big\{\frac{t}{\sqrt{n}} \sum_{i=1}^{n} \frac{X_i - \mu}{\sigma}\Big\} = \Big\{m\Big(\frac{t}{\sqrt{n}}\Big)\Big\}^n.$$

By Taylor's theorem, $m(s) = 1 + \frac{s^2}{2} + R(s)$ with $R(s)/s^2 \to 0$ as $s \to 0$. Hence

$$m\Big(\frac{t}{\sqrt{n}}\Big) = 1 + \frac{t^2}{2n} + R_n, \qquad nR_n = t^2 \frac{R(t/\sqrt{n})}{(t/\sqrt{n})^2} \to 0.$$

Using $\log(1 + x) = x + o(x)$ as $x \to 0$,

$$\log M_{\sqrt{n}(\bar{X}_n - \mu)/\sigma}(t) = n \log\Big(1 + \frac{t^2}{2n} + R_n\Big) = n\Big(\frac{t^2}{2n} + r_n\Big), \qquad nr_n \to 0,$$

so the mgf tends to $e^{t^2/2}$, the mgf of $N(0,1)$. The continuity theorem finishes the proof. ∎

In practice we read the theorem as $P(a < \bar{X}_n \le b) \approx \Phi(\frac{b-\mu}{\sigma/\sqrt{n}}) - \Phi(\frac{a-\mu}{\sigma/\sqrt{n}})$ for large n. The convergence of the cdfs is in fact uniform in x, which justifies replacing fixed a, b by values that depend on n.

Example 5.1 (Poisson and gamma)

(a) If $Y \sim \text{Poisson}(\lambda n)$, then $Y \overset{d}{=} X_1 + \cdots + X_n$ with $X_i \overset{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda)$, so $(Y - n\lambda)/\sqrt{n\lambda}$ is approximately $N(0,1)$. Thus a Poisson variable with a large mean ν is approximately $N(\nu, \nu)$. For $Y \sim \text{Poisson}(100)$, $P(90 < Y \le 110) \approx \Phi(1) - \Phi(-1) = 0.683$.
(b) Likewise $\text{Gamma}(n\alpha, \beta)$ is the law of a sum of n i.i.d. $\text{Gamma}(\alpha, \beta)$ variables, so for large shape it is approximately normal with the same mean and variance. In particular

$\chi^2(r) \approx N(r, 2r)$ for large r. For $Y \sim \chi^2(50)$, $P(Y > 60) \approx 1 - \Phi(10/10) = 0.159$, against the exact value 0.155.

The theorem extends to random vectors; the proof is the same, with the joint mgf in place of the mgf.

Theorem 5.2 (Multivariate central limit theorem)

Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be i.i.d. random vectors in $\mathbb{R}^k$ with mean μ and covariance matrix Σ. Then for every $\mathbf{x} \in \mathbb{R}^k$,

$$\lim_{n\to\infty} P\big(\sqrt{n}(\bar{\mathbf{X}}_n - \mu) \le \mathbf{x}\big) = P(\mathbf{Z} \le \mathbf{x}), \qquad \mathbf{Z} \sim N_k(\mathbf{0}, \Sigma),$$

where the inequalities are componentwise. (If Σ is singular, the statement holds at all $\mathbf{x}$ where the limit cdf is continuous.)

Example 5.2 (Multinomial)

Let $\mathbf{X}_n = (X_{n1}, \dots, X_{nk})' \sim \text{Multi}(n, \mathbf{p})$ with $\mathbf{p} = (p_1, \dots, p_k)'$. By the representation of Chapter 3, $\mathbf{X}_n \overset{d}{=} \mathbf{Z}_1 + \cdots + \mathbf{Z}_n$ with $\mathbf{Z}_i \overset{\text{i.i.d.}}{\sim} \text{Multi}(1, \mathbf{p})$, whose mean is $\mathbf{p}$ and covariance is $D(\mathbf{p}) - \mathbf{p}\mathbf{p}'$, where $D(\mathbf{p})$ is the diagonal matrix with entries p_j. Hence

$$\frac{\mathbf{X}_n - n\mathbf{p}}{\sqrt{n}} \text{ is approximately } N_k(\mathbf{0},\ D(\mathbf{p}) - \mathbf{p}\mathbf{p}').$$

This covariance matrix is singular: $(D(\mathbf{p}) - \mathbf{p}\mathbf{p}')\mathbf{1} = \mathbf{p} - \mathbf{p} = \mathbf{0}$, which reflects $\sum_j X_{nj} = n$.

5.2 Limiting distributions and convergence in probability

Motivation. Large-sample statistics become approximately normal or concentrate at a constant.

Example 5.3

Let $X_n \equiv 1/n$. Its cdf $F_n(x) = \mathbf{1}\{x \ge 1/n\}$ converges to $G(x) = \mathbf{1}\{x > 0\}$. G is not a cdf, but agrees with the cdf $F(x) = \mathbf{1}\{x \ge 0\}$ of 0 except at its discontinuity point: such points are ignored.

Definition (Limiting distribution)

Let X_n $(n = 1, 2, \dots)$ and Z be random variables, and let $C(F_Z)$ be the set of continuity points of F_Z. If

$$\lim_{n\to\infty} P(X_n \le x) = P(Z \le x) \qquad \text{for all } x \in C(F_Z),$$

then the distribution of Z is the **limiting** (or asymptotic) distribution of X_n, and X_n **converges in distribution** to Z, written $X_n \xrightarrow{d} Z$.

The CLT says $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} N(0, 1)$.

Example 5.4

(a) For $X_n \sim B(n, \lambda/n)$ and $X \sim \text{Poisson}(\lambda)$, Chapter 3 showed $P(X_n = k) \to P(X = k)$ for every k. Both are supported on $\{0, 1, 2, \dots\}$, so $P(X_n \le x) = \sum_{0 \le k \le x} P(X_n = k) \to P(X \le x)$ for all x (a finite sum). Hence $X_n \xrightarrow{d} \text{Poisson}(\lambda)$.
(b) Let $U_{(1)}$ be the minimum of n i.i.d. $U(0,1)$ variables. For $0 \le x \le n$,

$$P(nU_{(1)} > x) = \left(1 - \frac{x}{n}\right)^n \to e^{-x},$$

so $nU_{(1)} \xrightarrow{d} \text{Exp}(1)$. The same computation gives $n(1 - U_{(n)}) \xrightarrow{d} \text{Exp}(1)$.

Theorem 5.3 (Convergence to a constant)

Let c be a constant. Then $X_n \xrightarrow{d} c$ (i.e. to a variable X with $P(X = c) = 1$) if and only if

$$\lim_{n \to \infty} P(|X_n - c| \ge \epsilon) = 0 \qquad \text{for every } \epsilon > 0.$$

Proof. The cdf of the constant c is $\mathbf{1}\{x \ge c\}$, continuous everywhere except at c.

($\Rightarrow$) Fix $\epsilon > 0$. Since $c - \epsilon$ and $c + \epsilon/2$ are continuity points,

$$P(X_n \le c - \epsilon) \to 0, \qquad P(X_n \ge c + \epsilon) \le P(X_n > c + \epsilon/2) = 1 - P(X_n \le c + \epsilon/2) \to 0.$$

Adding, $P(|X_n - c| \ge \epsilon) \to 0$.

($\Leftarrow$) For $x < c$, put $\epsilon = c - x$: $0 \le P(X_n \le x) \le P(|X_n - c| \ge \epsilon) \to 0$. For $x > c$, put $\epsilon = x - c$: $0 \le 1 - P(X_n \le x) = P(X_n > x) \le P(|X_n - c| \ge \epsilon) \to 0$. So $P(X_n \le x) \to \mathbf{1}\{x \ge c\}$ for all $x \ne c$. ■

Definition (Convergence in probability)

X_n **converges in probability** to the constant c, written $X_n \xrightarrow{p} c$ or $\operatorname{plim} X_n = c$, if $\lim_n P(|X_n - c| \ge \epsilon) = 0$ for every $\epsilon > 0$. For random vectors, $\mathbf{X}_n \xrightarrow{p} \mathbf{c}$ means $P(\|\mathbf{X}_n - \mathbf{c}\| \ge \epsilon) \to 0$ for every $\epsilon > 0$, with $\|\cdot\|$ the Euclidean norm.

By Theorem 5.3, convergence in probability to a constant is the same as convergence in distribution to that constant. An estimator that converges in probability to its target is called **consistent**.

Theorem 5.4 (Weak law of large numbers)

If $X_1, \dots, X_n$ are i.i.d. and $EX_1 = \mu$ is finite, then $\bar{X}_n \xrightarrow{p} \mu$.

Proof. The general proof is beyond our level; we assume in addition $\sigma^2 = \operatorname{Var} X_1 < \infty$. By Chebyshev's inequality (Theorem 1.10), for every $\epsilon > 0$,

$$0 \le P(|\bar{X}_n - \mu| \ge \epsilon) \le \frac{\operatorname{Var} \bar{X}_n}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2} \to 0.$$

■

Applied to indicators $\mathbf{1}_A(X_i)$, the law says that the relative frequency of an event converges in probability to its probability:

$$\frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_A(X_i) \xrightarrow{p} E\mathbf{1}_A(X_1) = P(X_1 \in A).$$

This is the frequency interpretation of probability, and the reason a sample mean is a sensible estimate of a population mean.

Theorem 5.5

For random vectors $\mathbf{X}_n = (X_{n1}, \ldots, X_{nk})'$ and $\mathbf{c} = (c_1, \ldots, c_k)'$, the following are equivalent:

(a) $\mathbf{X}_n \xrightarrow{p} \mathbf{c}$;

(b) $P(\max_{1\le i\le k}|X_{ni} - c_i| \ge \epsilon) \to 0$ for every $\epsilon > 0$;

(c) $X_{ni} \xrightarrow{p} c_i$ for each $i = 1, \ldots, k$.

Proof. (a)$\Leftrightarrow$(b): From $\max_i|x_i| \le \|\mathbf{x}\| \le \sqrt{k}\max_i|x_i|$,

$$P(\max_i|X_{ni} - c_i| \ge \epsilon) \le P(\|\mathbf{X}_n - \mathbf{c}\| \ge \epsilon) \le P(\max_i|X_{ni} - c_i| \ge \epsilon/\sqrt{k}).$$

(b)$\Leftrightarrow$(c): Since $\{|X_{ni} - c_i| \ge \epsilon\} \subseteq \{\max_j|X_{nj} - c_j| \ge \epsilon\} = \bigcup_j\{|X_{nj} - c_j| \ge \epsilon\}$,

$$\max_i P(|X_{ni} - c_i| \ge \epsilon) \le P(\max_i|X_{ni} - c_i| \ge \epsilon) \le \sum_{i=1}^{k} P(|X_{ni} - c_i| \ge \epsilon).$$

■

Corollary 5.6 (Multivariate law of large numbers)

If $\mathbf{X}_1, \ldots, \mathbf{X}_n$ are i.i.d. random vectors with finite mean μ, then $\bar{\mathbf{X}}_n \xrightarrow{p} \mu$.

Proof. Apply Theorem 5.4 to each coordinate and use Theorem 5.5. ■

Example 5.5 (Sample moments)

The r-th **sample moment** $\hat{m}_r = \frac{1}{n}\sum X_i^r$ estimates the population moment $m_r = EX_1^r$. If $E|X_1|^k < \infty$, all lower moments are finite too, and Corollary 5.6 applied to $(X_i, X_i^2, \ldots, X_i^k)$ gives

$$(\hat{m}_1, \ldots, \hat{m}_k)' \xrightarrow{p} (m_1, \ldots, m_k)'.$$

Theorem 5.7 (Continuous functions preserve convergence in probability)

If $\mathbf{X}_n \xrightarrow{p} \mathbf{c}$ and g is continuous at $\mathbf{c}$, then $g(\mathbf{X}_n) \xrightarrow{p} g(\mathbf{c})$.

Proof. Given $\epsilon > 0$, choose $\delta > 0$ such that $\|\mathbf{x} - \mathbf{c}\| < \delta$ implies $|g(\mathbf{x}) - g(\mathbf{c})| < \epsilon$. Then $\{|g(\mathbf{X}_n) - g(\mathbf{c})| \ge \epsilon\} \subseteq \{\|\mathbf{X}_n - \mathbf{c}\| \ge \delta\}$, so

$$P(|g(\mathbf{X}_n) - g(\mathbf{c})| \ge \epsilon) \le P(\|\mathbf{X}_n - \mathbf{c}\| \ge \delta) \to 0.$$

■

Corollary 5.8

If $X_n \xrightarrow{p} a$ and $Y_n \xrightarrow{p} b$, then $X_n \pm Y_n \xrightarrow{p} a \pm b$, $X_nY_n \xrightarrow{p} ab$, and $X_n/Y_n \xrightarrow{p} a/b$ when $b \ne 0$.

Proof. By Theorem 5.5, $(X_n, Y_n)' \xrightarrow{p} (a, b)'$. The functions $x \pm y$, xy and x/y (for $y \ne 0$) are continuous at (a, b); apply Theorem 5.7. ■

Example 5.6 (Consistency of S_n^2 and S_n)

Let $X_1, \ldots, X_n$ be i.i.d. with $EX_1^2 < \infty$ and $\sigma^2 = \operatorname{Var} X_1 > 0$. The sample variance is a function of the first two sample moments:

$$S_n^2 = \frac{1}{n-1}\sum (X_i - \bar{X})^2 = \frac{n}{n-1}(\hat{m}_2 - \hat{m}_1^2).$$

Since $\hat{m}_1 \xrightarrow{p} m_1$, $\hat{m}_2 \xrightarrow{p} m_2$ and $n/(n-1) \to 1$, Corollary 5.8 gives $S_n^2 \xrightarrow{p} m_2 - m_1^2 = \sigma^2$. Since $\sqrt{x}$ is continuous, $S_n \xrightarrow{p} \sigma$ as well.

Theorem 5.9

If $\operatorname{Var} X_n < \infty$, $EX_n \to a$ and $\operatorname{Var} X_n \to 0$, then $X_n \xrightarrow{p} a$.

Proof. By Markov's inequality applied to $(X_n - a)^2$,

$$P(|X_n - a| \geq \epsilon) \leq \frac{E(X_n - a)^2}{\epsilon^2} = \frac{\operatorname{Var} X_n + (EX_n - a)^2}{\epsilon^2} \to 0.$$

■

Example 5.7

For uniform order statistics, $U_{(1)} \sim \text{Beta}(1, n)$, so $EU_{(1)} = 1/(n+1) \to 0$ and $\operatorname{Var} U_{(1)} = n/\{(n+1)^2(n+2)\} \to 0$. Hence $U_{(1)} \xrightarrow{p} 0$, and similarly $U_{(n)} \xrightarrow{p} 1$. More generally, for a sample from $f(x;\theta) = e^{-(x-\theta)}$, $x > \theta$, the minimum $X_{(1)} = \theta + E_{(1)}$ with $E_{(1)} \sim \text{Exp}(1/n)$, so $X_{(1)} \xrightarrow{p} \theta$.

Example 5.8 (Consistency of sample quantiles)

Let F be continuous with inverse F^{-1}, fix $0 < \alpha < 1$, and let r_n be integers with $r_n/n \to \alpha$. The order statistic $X_{(r_n)}$ is a **sample α-quantile**. We show $X_{(r_n)} \xrightarrow{p} F^{-1}(\alpha)$ when F^{-1} is continuous at α.
By Corollary 4.14 and Example 4.12, with $Z_1, \ldots, Z_n \overset{\text{i.i.d.}}{\sim} \text{Exp}(1)$,

$$X_{(r_n)} \overset{d}{=} h(Y_n), \qquad Y_n = \frac{Z_1}{n} + \frac{Z_2}{n-1} + \cdots + \frac{Z_{r_n}}{n - r_n + 1}, \qquad h(y) = F^{-1}(1 - e^{-y}).$$

Indeed $1 - e^{-E}$ is uniform for $E \sim \text{Exp}(1)$, so $U_{(r)} \overset{d}{=} 1 - e^{-E_{(r)}}$. Now, by Riemann sums,

$$EY_n = \sum_{i=1}^{r_n} \frac{1}{n-i+1} = \frac{1}{n}\sum_{k=0}^{r_n - 1} \frac{1}{1 - k/n} \to \int_0^\alpha \frac{dx}{1-x} = -\log(1-\alpha),$$

$$\operatorname{Var} Y_n = \sum_{i=1}^{r_n} \frac{1}{(n-i+1)^2} = \frac{1}{n} \cdot \frac{1}{n}\sum_{k=0}^{r_n - 1} \frac{1}{(1-k/n)^2} \approx \frac{1}{n} \cdot \frac{\alpha}{1-\alpha} \to 0.$$

By Theorem 5.9, $Y_n \xrightarrow{p} -\log(1-\alpha)$. Since h is continuous at $-\log(1-\alpha)$, Theorem 5.7 gives $h(Y_n) \xrightarrow{p} h(-\log(1-\alpha)) = F^{-1}(\alpha)$. Convergence in probability to a constant depends only on the distributions, so $X_{(r_n)} \xrightarrow{p} F^{-1}(\alpha)$.

5.3 Computing limiting distributions

Motivation. CLT, Slutsky, continuous mapping and the delta method handle almost any statistic of sample means.

Theorem 5.10 (Slutsky)

If $X_n \xrightarrow{d} Z$ and $Y_n \xrightarrow{p} c$ for a constant c, then

(a) $X_n + Y_n \xrightarrow{d} Z + c$;

(b) $X_n - Y_n \xrightarrow{d} Z - c$;

(c) $Y_n X_n \xrightarrow{d} cZ$;

(d) $X_n / Y_n \xrightarrow{d} Z/c$ if $c \neq 0$.

Proof. The theorem is true in general; we prove it when F_Z is continuous, which covers our applications.

(a) For $\epsilon > 0$,

$$P(X_n+Y_n \le z) \le P(X_n+Y_n \le z,\ |Y_n-c| < \epsilon)+P(|Y_n-c| \ge \epsilon) \le P(X_n \le z-c+\epsilon)+P(|Y_n-c| \ge \epsilon).$$

Letting $n \to \infty$, $\limsup_n P(X_n + Y_n \le z) \le P(Z \le z - c + \epsilon)$. In the same way,

$$P(X_n + Y_n > z) \le P(X_n > z - c - \epsilon) + P(|Y_n - c| \ge \epsilon),$$

so $\liminf_n P(X_n + Y_n \le z) \ge P(Z \le z - c - \epsilon)$. Let $\epsilon \downarrow 0$; by continuity of F_Z both bounds tend to $P(Z + c \le z)$.

(b) $-Y_n \xrightarrow{p} -c$; apply (a).

(c) First let $c = 0$. For $\epsilon > 0$ and any $k > 0$,

$$P(|Y_n X_n| \ge \epsilon) \le P(|X_n| > k) + P(|Y_n| \ge \epsilon/k).$$

As $n \to \infty$ the right side tends to $P(Z > k) + P(Z < -k)$ (at continuity points), and this is arbitrarily small for large k. So $Y_n X_n \xrightarrow{p} 0$. For general c, write $Y_n X_n = (Y_n - c)X_n + cX_n$: the first term $\xrightarrow{p} 0$ by the case just proved, and $cX_n \xrightarrow{d} cZ$ (obvious for $c \neq 0$, trivial for $c = 0$). Then apply (a).

(d) $1/Y_n \xrightarrow{p} 1/c$ by Theorem 5.7; apply (c). ■

Example 5.9 (Studentized mean)

Let $X_1, \dots, X_n$ be i.i.d. with mean μ and variance $\sigma^2 \in (0, \infty)$. The **studentized mean** satisfies

$$\frac{\bar{X}_n - \mu}{S_n/\sqrt{n}} = \frac{\sqrt{n}(\bar{X}_n - \mu)/\sigma}{S_n/\sigma} \xrightarrow{d} \frac{Z}{1} = Z \sim N(0,1),$$

by the CLT, Example 5.6 and Slutsky. Two consequences:

- For normal samples the left side is exactly $t(n-1)$, so $t(n-1) \xrightarrow{d} N(0,1)$: for large n, t quantiles are close to normal ones.
- For any population with finite variance, $P(\bar{X}_n - z_{\alpha/2} S_n/\sqrt{n} \le \mu \le \bar{X}_n + z_{\alpha/2} S_n/\sqrt{n}) \to 1 - \alpha$, so $\bar{X}_n \pm z_{\alpha/2} S_n/\sqrt{n}$ is an **asymptotic** $100(1-\alpha)\%$ confidence interval for μ.

For instance, if $n = 100$ waiting times give $\bar{x} = 12.4$ and $s = 3.0$, the asymptotic 95% interval is $12.4 \pm 1.96 \cdot 0.3 = (11.81, 12.99)$, whatever the shape of the population.

Example 5.10 (Sample variance)

Let $\mu_4 = E(X_1 - \mu)^4 < \infty$. Write

$$S_n^2 = \frac{n}{n-1}\Big\{\frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2 - (\bar{X}_n - \mu)^2\Big\}.$$

The variables $Y_i = (X_i - \mu)^2$ are i.i.d. with mean σ^2 and variance $\mu_4 - \sigma^4$, so by the CLT $\sqrt{n}(\frac{1}{n}\sum Y_i - \sigma^2) \xrightarrow{d} N(0, \mu_4 - \sigma^4)$. Also $\sqrt{n}(\bar{X}_n - \mu)^2 = \sqrt{n}(\bar{X}_n - \mu) \cdot (\bar{X}_n - \mu) \xrightarrow{d} 0$ by Slutsky (a $\xrightarrow{d} N(0, \sigma^2)$ factor times a $\xrightarrow{p} 0$ factor). Therefore

$$\sqrt{n}(S_n^2 - \sigma^2) = \frac{n}{n-1}\Big[\sqrt{n}\Big(\frac{1}{n}\sum Y_i - \sigma^2\Big) - \sqrt{n}(\bar{X}_n - \mu)^2\Big] + \frac{\sqrt{n}}{n-1}\sigma^2 \xrightarrow{d} N(0, \mu_4 - \sigma^4).$$

With the excess kurtosis $\rho_4 = \mu_4/\sigma^4 - 3$, the limit variance is $(\rho_4 + 2)\sigma^4$. For normal populations $\rho_4 = 0$ and the variance is $2\sigma^4$, in line with $\operatorname{Var} S_n^2 = 2\sigma^4/(n-1)$.

Definition (Convergence in distribution of vectors)

$\mathbf{X}_n \xrightarrow{d} \mathbf{Z}$ in $\mathbb{R}^k$ if $P(\mathbf{X}_n \le \mathbf{x}) \to P(\mathbf{Z} \le \mathbf{x})$ at every $\mathbf{x}$ where the joint cdf of $\mathbf{Z}$ is continuous.

Theorem 5.11 (Continuous mapping)

If $\mathbf{X}_n \xrightarrow{d} \mathbf{Z}$ in $\mathbb{R}^k$ and g is continuous, then $g(\mathbf{X}_n) \xrightarrow{d} g(\mathbf{Z})$.

(Proof omitted.) Slutsky follows only with joint convergence of (X_n, Y_n), which holds when one component tends to a constant.

Example 5.11 (Binomial and chi-square)

If $X_n \sim B(n, p)$, then $(X_n - np)/\sqrt{np(1-p)} \xrightarrow{d} Z$, and squaring (continuous) gives $\frac{(X_n - np)^2}{np(1-p)} \xrightarrow{d} \chi^2(1)$. With two cells $X_{n1} = X_n$, $X_{n2} = n - X_n$ and $p_1 = p$, $p_2 = 1 - p$,

$$\frac{(X_{n1} - np_1)^2}{np_1} + \frac{(X_{n2} - np_2)^2}{np_2} = (X_n - np)^2\Big(\frac{1}{np} + \frac{1}{n(1-p)}\Big) = \frac{(X_n - np)^2}{np(1-p)}.$$

So the two-cell Pearson statistic is asymptotically $\chi^2(1)$.

Example 5.12 (Pearson's chi-square statistic)

Let $(X_{n1}, \dots, X_{nk})' \sim \text{Multi}(n, (p_1, \dots, p_k)')$, all $p_j > 0$. We show

$$Q_n = \sum_{j=1}^{k} \frac{(X_{nj} - np_j)^2}{np_j} \xrightarrow{d} \chi^2(k-1).$$

Drop the last cell: let $\mathbf{Y}_n = (X_{n1}, \dots, X_{n,k-1})'$, $\mu = (p_1, \dots, p_{k-1})'$ and $r = k - 1$. By Example 5.2, $(\mathbf{Y}_n - n\mu)/\sqrt{n} \xrightarrow{d} \mathbf{Z} \sim N_r(\mathbf{0}, \Sigma)$ with $\Sigma = D(\mu) - \mu\mu'$, which is now nonsingular.

The quadratic form $\mathbf{z}'\Sigma^{-1}\mathbf{z}$ is continuous, so by Theorem 5.11 and Theorem 4.18(a),

$$\frac{1}{n}(\mathbf{Y}_n - n\mu)'\Sigma^{-1}(\mathbf{Y}_n - n\mu) \xrightarrow{d} \mathbf{Z}'\Sigma^{-1}\mathbf{Z} \sim \chi^2(k-1).$$

It remains to identify the left side with Q_n. One checks directly that

$$\Sigma^{-1} = D(\mu)^{-1} + \frac{1}{p_k}\mathbf{1}\mathbf{1}' :$$

multiplying, $(D - \mu\mu')(D^{-1} + \mathbf{1}\mathbf{1}'/p_k) = I + \mu\mathbf{1}'/p_k - \mu\mathbf{1}' - (1-p_k)\mu\mathbf{1}'/p_k = I$, using $D\mathbf{1} = \mu$ and $\mu'\mathbf{1} = 1 - p_k$. Hence, with $d_j = X_{nj} - np_j$ and $\sum_{j<k} d_j = -d_k$,

$$\frac{1}{n}(\mathbf{Y}_n - n\mu)'\Sigma^{-1}(\mathbf{Y}_n - n\mu) = \sum_{j=1}^{k-1}\frac{d_j^2}{np_j} + \frac{(\sum_{j<k} d_j)^2}{np_k} = \sum_{j=1}^{k}\frac{d_j^2}{np_j} = Q_n.$$

Theorem 5.12 (Delta method)

Suppose $\sqrt{n}(\mathbf{X}_n - \theta) \xrightarrow{d} \mathbf{Z}$ in $\mathbb{R}^k$, and $g : \mathbb{R}^k \to \mathbb{R}$ has continuous first partial derivatives near θ, with gradient $\dot{g}(\theta) = (\partial g/\partial\theta_1, \dots, \partial g/\partial\theta_k)'$. Then

$$\sqrt{n}(g(\mathbf{X}_n) - g(\theta)) \xrightarrow{d} \dot{g}(\theta)'\mathbf{Z}.$$

In particular, if $\mathbf{Z} \sim N_k(\mathbf{0}, \Sigma)$, the limit is $N\big(0, \dot{g}(\theta)'\Sigma\dot{g}(\theta)\big)$.

Proof. We give the one-dimensional proof; the multivariate one is the same. Since g is differentiable at θ, there is a function r with $r(x) \to 0$ as $x \to \theta$ such that

$$g(x) = g(\theta) + \{\dot{g}(\theta) + r(x)\}(x - \theta).$$

Set $r(\theta) = 0$. First, $X_n \xrightarrow{p} \theta$: by Slutsky, $X_n - \theta = \frac{1}{\sqrt{n}} \cdot \sqrt{n}(X_n - \theta) \xrightarrow{d} 0 \cdot Z = 0$, and Theorem 5.3 applies. Next, $r(X_n) \xrightarrow{p} 0$: given $\epsilon > 0$ choose δ with $|x - \theta| < \delta \Rightarrow |r(x)| < \epsilon$; then $P(|r(X_n)| \geq \epsilon) \leq P(|X_n - \theta| \geq \delta) \to 0$. Finally

$$\sqrt{n}(g(X_n) - g(\theta)) = \{\dot{g}(\theta) + r(X_n)\}\sqrt{n}(X_n - \theta),$$

where the first factor $\xrightarrow{p} \dot{g}(\theta)$ and the second $\xrightarrow{d} Z$. Slutsky's theorem gives the limit $\dot{g}(\theta)Z$. ∎

Example 5.13 (Sample standard deviation)

From Example 5.10, $\sqrt{n}(S_n^2 - \sigma^2) \xrightarrow{d} N(0, (\rho_4 + 2)\sigma^4)$. With $g(x) = \sqrt{x}$, $\dot{g}(\sigma^2) = 1/(2\sigma)$, so

$$\sqrt{n}(S_n - \sigma) \xrightarrow{d} N\Big(0, \frac{(\rho_4 + 2)\sigma^2}{4}\Big).$$

For normal data this is $N(0, \sigma^2/2)$.

Example 5.14 (Sample correlation coefficient)

Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be i.i.d. pairs with correlation ρ and finite fourth moments, and

$$\hat\rho_n = \frac{\sum (X_i - \bar X)(Y_i - \bar Y)}{\sqrt{\sum (X_i - \bar X)^2}\sqrt{\sum (Y_i - \bar Y)^2}}.$$

$\hat\rho_n$ is unchanged if we standardize each coordinate, so we may assume $\mu_1 = \mu_2 = 0$, $\sigma_1 = \sigma_2 = 1$. Write the five sample means $\bar X, \bar Y, \overline{XY}, \overline{X^2}, \overline{Y^2}$ as the average of $\mathbf{W}_i = (X_i, Y_i, X_iY_i, X_i^2, Y_i^2)'$. Then $\hat\rho_n = g(\bar{\mathbf{W}}_n)$ with

$$g(t_1, \dots, t_5) = \frac{t_3 - t_1 t_2}{\sqrt{t_4 - t_1^2}\sqrt{t_5 - t_2^2}}, \qquad \theta = E\mathbf{W}_1 = (0, 0, \rho, 1, 1)', \qquad g(\theta) = \rho.$$

The gradient at θ is $\dot g(\theta) = (0, 0, 1, -\rho/2, -\rho/2)'$. By the multivariate CLT and Theorem 5.12,

$$\sqrt{n}(\hat\rho_n - \rho) \xrightarrow{d} N\Big(0,\ \operatorname{Var}\Big(X_1Y_1 - \frac{\rho}{2}X_1^2 - \frac{\rho}{2}Y_1^2\Big)\Big).$$

Bivariate normal case. Write $Y_1 = \rho X_1 + \sqrt{1-\rho^2}\,T$ with X_1, T independent $N(0,1)$ (Theorem 4.17). Expanding,

$$X_1Y_1 - \frac{\rho}{2}X_1^2 - \frac{\rho}{2}Y_1^2 = (1-\rho^2)\Big\{\frac{\rho}{2}(X_1^2 - T^2) + \sqrt{1-\rho^2}\,X_1T\Big\}.$$

Here $\operatorname{Var}(X_1^2 - T^2) = 4$, $\operatorname{Var}(X_1T) = 1$, and the two terms are uncorrelated. So the variance is $(1-\rho^2)^2\{\rho^2 + 1 - \rho^2\} = (1-\rho^2)^2$:

$$\sqrt{n}(\hat\rho_n - \rho) \xrightarrow{d} N(0, (1-\rho^2)^2).$$

Example 5.15 (Variance-stabilizing transformation)

In Example 5.14 the limit variance depends on the unknown ρ. For a function g, the delta method gives $\sqrt{n}(g(\hat\rho_n) - g(\rho)) \xrightarrow{d} N(0, \dot g(\rho)^2(1-\rho^2)^2)$. Choosing $\dot g(\rho) = 1/(1-\rho^2)$, i.e. **Fisher's z-transform**

$$g(\rho) = \frac{1}{2}\log\frac{1+\rho}{1-\rho} = \tanh^{-1}\rho,$$

makes the limit $N(0,1)$, free of ρ. Such a g is a **variance-stabilizing transformation**. The asymptotic interval $\tanh^{-1}\hat\rho_n \pm z_{\alpha/2}/\sqrt{n}$ for $\tanh^{-1}\rho$ maps back to an interval for ρ. With $n = 100$ and $\hat\rho = 0.6$: $\tanh^{-1}0.6 = 0.693$, the interval is $0.693 \pm 0.196 = (0.497, 0.889)$, and applying tanh gives $(0.460, 0.711)$ for ρ.

Example 5.16 (Asymptotic normality of sample quantiles)

In the setting of Example 5.8, assume F has a density f with $f(\xi_\alpha) > 0$ at $\xi_\alpha = F^{-1}(\alpha)$, and $r_n/n = \alpha + o(n^{-1/2})$. Then

$$\sqrt{n}(X_{(r_n)} - \xi_\alpha) \xrightarrow{d} N\Big(0, \frac{\alpha(1-\alpha)}{f(\xi_\alpha)^2}\Big).$$

Sketch. With Y_n as in Example 5.8, its cgf is $K(s) = -\sum_{i=1}^{r_n}\log\big(1 - \frac{s}{n-i+1}\big) = \sum_{m\ge1}\frac{s^m}{m}\sum_i (n-i+1)^{-m}$. Taking $s = \sqrt{n}\,t/\tau$ with $\tau^2 = \alpha/(1-\alpha)$, the $m = 1$ term gives the centring, the $m = 2$ term tends to $t^2/2$, and the terms with $m \ge 3$ are

$O(n^{m/2} \cdot n^{1-m}) \to 0$. Hence $\sqrt{n}(Y_n + \log(1-\alpha)) \xrightarrow{d} N(0, \alpha/(1-\alpha))$. Next apply the delta method to $h(y) = F^{-1}(1-e^{-y})$, whose derivative at $y_0 = -\log(1-\alpha)$ is

$$\dot{h}(y_0) = \frac{e^{-y_0}}{f(F^{-1}(1-e^{-y_0}))} = \frac{1-\alpha}{f(\xi_\alpha)}.$$

The limit variance is $\frac{(1-\alpha)^2}{f(\xi_\alpha)^2} \cdot \frac{\alpha}{1-\alpha} = \frac{\alpha(1-\alpha)}{f(\xi_\alpha)^2}$.
For the sample median ($\alpha = 1/2$), the variance is $1/\{4f(m)^2\}$. For $N(\mu, \sigma^2)$ data, $f(\mu) = 1/(\sigma\sqrt{2\pi})$, so the median has asymptotic variance $\pi\sigma^2/2$, against σ^2 for the mean: the mean is more efficient by the factor $\pi/2 \approx 1.57$. For double exponential data $\frac{1}{2}e^{-|x-\mu|}$, the median has variance 1 against 2 for the mean, and the order is reversed.

5.4 Approximation by simulation

Computer imitations of observations are **random numbers** (**uniform random numbers** for $U(0,1)$); since $F^{-1}(U) \sim F$ (Theorem 4.13), uniform ones give any distribution.

Example 5.17 (Generating random numbers)

- Logistic: $F(z) = 1/(1+e^{-z})$ has $F^{-1}(u) = \log\frac{u}{1-u}$, so $\log\frac{U}{1-U} \sim L(0,1)$, and $\mu + \sigma\log\frac{U}{1-U} \sim L(\mu,\sigma)$.
- Exponential: $-\beta\log(1-U) \sim \text{Exp}(\beta)$ (and so is $-\beta\log U$).
- Cauchy: $\tan\{\pi(U-\frac{1}{2})\}$ is standard Cauchy.
- Normal: $\mu + \sigma\Phi^{-1}(U) \sim N(\mu,\sigma^2)$; alternatively use the polar method of Example 4.6.

Sums and ratios of such variates give gamma, χ^2, t and F random numbers through the representations of Chapter 4.

Theorem 5.13 (Monte Carlo integration)

Let g be continuous on $[a,b]$ and $U_1, \ldots, U_n \overset{\text{i.i.d.}}{\sim} U(0,1)$. With $X_i = a + (b-a)U_i \sim U(a,b)$,

$$\frac{b-a}{n}\sum_{i=1}^{n} g(X_i) \xrightarrow{p} \int_a^b g(x)\,dx.$$

Proof. X_i has pdf $\frac{1}{b-a}$ on (a,b), so $Eg(X_1) = \frac{1}{b-a}\int_a^b g(x)\,dx$, finite since g is bounded. By the law of large numbers, $\frac{b-a}{n}\sum g(X_i) \xrightarrow{p} (b-a)Eg(X_1) = \int_a^b g$. ■

So with uniform random numbers $u_1, \ldots, u_n$ and $y_i = (b-a)g(a+(b-a)u_i)$, the average $\bar{y}$ approximates the integral. Since $\bar{y}$ is a sample mean, Example 5.9 gives the asymptotic 95% interval $\bar{y} \pm 1.96 s_y/\sqrt{n}$ as an error bound; the error shrinks like $1/\sqrt{n}$.

Example 5.18

To approximate $\int_0^2 x^3\,dx = 4$, use $y_i = 2(2u_i)^3 = 16u_i^3$. Here $\text{Var}(16U^3) = 256(\frac{1}{7} - \frac{1}{16}) = 20.6$, so with $n = 10{,}000$ the standard error is about $\sqrt{20.6}/100 = 0.045$, and the 95% error bound is about ± 0.09. Quadrupling n halves it.

Example 5.19 (Variance of the logistic distribution)

The standard logistic distribution is symmetric about 0 with variance $\int x^2 e^{-x}(1+e^{-x})^{-2}dx = \pi^2/3 \approx 3.290$. (Expand $(1+e^{-x})^{-2} = \sum_{m\geq 1}(-1)^{m-1}me^{-(m-1)x}$ on $x > 0$ to get $4\sum_{m\geq 1}(-1)^{m-1}/m^2 = 4\cdot\pi^2/12$.) To check this by simulation, take $x_i = \log\frac{u_i}{1-u_i}$ and average $y_i = x_i^2$; the estimate with its interval settles near 3.29 as n grows.

Example 5.20 (Estimating π)

If $U_1, U_2 \overset{\text{i.i.d.}}{\sim} U(0,1)$, the point (U_1, U_2) is uniform on the unit square and $p = P(U_1^2 + U_2^2 \leq 1) = \pi/4$. With n independent points, let $\hat{p}_n$ be the proportion inside the quarter disk. By the LLN, $4\hat{p}_n \xrightarrow{p} \pi$, and by the CLT and Slutsky,

$$\frac{\hat{p}_n - p}{\sqrt{\hat{p}_n(1-\hat{p}_n)/n}} \xrightarrow{d} N(0,1),$$

so $4\hat{p}_n \pm 4z_{\alpha/2}\sqrt{\hat{p}_n(1-\hat{p}_n)/n}$ is an asymptotic interval for π. Since $4\sqrt{p(1-p)} \approx 1.64$, with $n = 10{,}000$ the 95% half-width is about $1.96\cdot 1.64/100 \approx 0.032$: two correct decimals need about 10^4 points, and each further digit costs a factor of 100.